Step-by-step manual for APEX Pipeline in OT-2

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Getting Started

This manual provides all the necessary information to set up and operate the Automated Protein EXpression (APEX) pipeline using the OT-2. In *Escherichia coli* (*E. coli*), a standard protein expression workflow typically consists of four sequential steps. We've designed APEX with a modular architecture that mirrors this process, implementing four self-contained modules, each describing a specific OT-2 protocol.

Protocol 1 Heat shock transformation to deliver plasmid DNA into competent cells.

Protocol 2 Antibiotic selection on agar plates to isolate colonies carrying the DNA payload.

Protocol 3 Colony sampling and culturing of successfully transformed cells.

Protocol 4 Chemical induction of protein expression.

Additionally, we have developed supplementary protocols for:

Protocol 5 Colony PCR sample preparation for rapid screening of successful transformants.

Protocol 6 SDS-PAGE sample preparation for protein expression analysis and verification.

Protocol 7 Optical Density at 600 nm (OD_{600}) measurement sample preparation for monitoring bacterial growth.

APEX operates through a structured configuration system requiring two key input files: a JavaScript Object Notation (JSON) file for OT-2 hardware settings and a Comma Separated Value (CSV) file containing experiment-specific liquid handling instructions. These files provide the essential parameters for laboratory protocol setup and execution.

The system integrates these inputs into a master Python script that can be directly imported into the robot via the Opentrons app. Simultaneously, APEX generates a PDF document featuring both textual instructions and visual representations of the labware layout, serving as a practical reference guide for users setting up experiments (Figure 1).

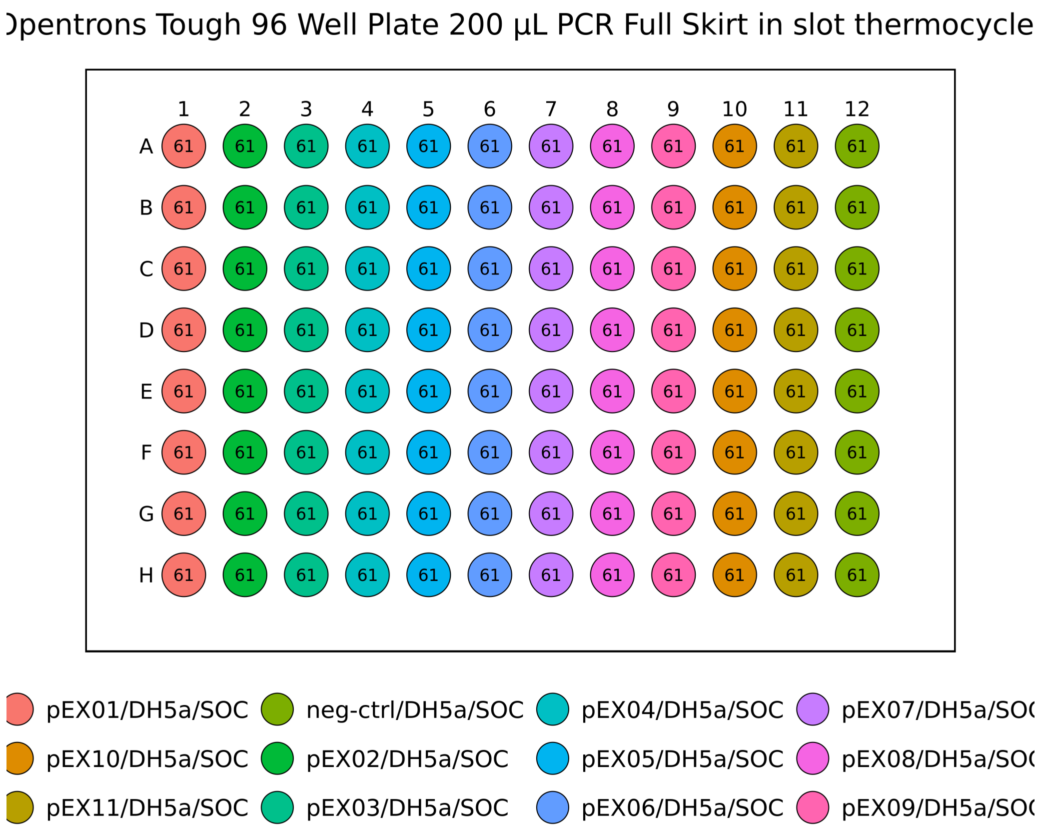

Figure 1: Example of labware visualised by APEX.

For validation purposes, APEX includes simulates each protocol that produces a detailed execution log, similar to the run log in the Opentrons App. This simulation output allows to preview and verify all protocol steps before physical implementation.

The pipeline is managed using Nextflow, a workflow manager that streamlines the execution of complex computational processes. Furthermore, APEX is fully containerised using Docker, which encloses all software dependencies into a single package. This containerisation ensures consistent deployment across any computational environment without requiring additional software installations.

System Requirements

Installing Java

Ensure Java 8 or later is installed on your system. Visit `https://www.java.com/en/download/` for installation instructions if needed. Java is required to run Nextflow.

Installing Nextflow

Open a terminal and execute the following command to download and install Nextflow:

```
curl -s https://get.nextflow.io | bash
```

Move the Nextflow executable to a directory in your PATH for easier access:

```
sudo mv nextflow /usr/local/bin
```

Verify the installation:

```
nextflow -version
```

Downloading and Running the Pipeline

The APEX pipeline is available on GitHub. Use either of the following commands to download it:

```
    nextflow pull stracquadaniolab/apex-nf
```

Or directly with the GitHub URL:

```
    nextflow pull https://github.com/stracquadaniolab/apex-nf
```

The pipeline will be downloaded to:

```
    $HOME/.nextflow/assets/stracquadaniolab/apex-nf
```

To run the complete pipeline:

```
    nextflow run stracquadaniolab/apex-nf
```

To run a specific protocol (e.g., Protocol 1):

```
    nextflow run stracquadaniolab/apex-nf -entry PROTOCOL_1
```

To view project details, installation location or check for updates:

```
    nextflow info stracquadaniolab/apex-nf
```

Configuration and Setup

The pipeline is organised in the "assets" folder with the following structure:

testdata JSON and CSV example files for protocol configurations and experimental data.

instructions R markdown template files containing experiment setup instructions.

labware JSON files defining supported labware. Custom labware definitions can be added here (more information here).

protocols Python template files for each protocol.

results Output files including ready-to-use Python protocols and PDF instructions.

Input Files

After downloading the necessary software, setting up your experiment with APEX is straightforward. Create a JSON and a CSV file containing all the details for your experiment. These files are critical as they dictate the setup and execution of your protocol.
Example files can be found in the `testdata` folder of the APEX directory once it has been downloaded. These examples serve as templates for proper formatting and contain all necessary parameters. The input files follow these naming conventions:

- **JSON File**: Name your file as `protocol-*-config.json`, where * represents the protocol ID number. For example, a transformation experiment would use `protocol-1-config.json`.
- **CSV File**: Similarly, name your CSV file as `protocol-*-data.csv`. For transformation, this would be `protocol-1-data.csv`.

Understanding JSON Configuration Files

JavaScript Object Notation (JSON) configuration file is used in the pipeline for setting up the Opentrons. This file includes the details about the hardware, such as types of pipettes available and layout of the Opentrons deck, specifying where each piece of labware should be placed.

Here's an example JSON configuration file that outlines a simple setup for an OT-2 robot:

```
{
    "pipette_name": "p20_multi_gen2",
    "source_plate_name": "armadillo_96_wellplate_200ul_pcr_full_skirt",
    "source_plate_slot": 4
}
```

For each protocol within the pipeline, this manual provides a specific JSON template with detailed explanations of all parameters.
To create a JSON file, you can use for example a simple text editor like Notepad (Windows) or TextEdit (Mac, set to plain text mode). Type your JSON data and save the file with a '.json'' extension by selecting Save As from the File menu. Move the saved file to the `testdata` folder within the APEX project directory.

Understanding CSV Data Files

Comma Separated Value (CSV) files provide detailed instructions for operations like liquid transfers within the protocol. They allow users to specify parameters such as transfer volumes, source and destination wells, and other experimental variables without modifying the underlying code.

Example of a CSV file:

```
source_well,destination_well,volume
A1,A2,50
B1,C2,25
```

For each protocol within the pipeline, this manual provides a specific CSV template with detailed explanations of all parameters.

To create a CSV file, use either a basic text editor or a spreadsheet program like Microsoft Excel. Enter your data, ensuring each field is separated by a comma without spaces. Save the file with a '.csv' extension, selecting 'CSV (Comma delimited) (*.csv)' if using a spreadsheet program. Ensure the file is saved in the `testdata` folder within the APEX project directory.

Output files

After running the pipeline, all outputs will be saved in the `results` folder. Output files follow these naming conventions:

- **Python protocol**: `protocol-*.py`, where * represents the ID number of the protocol. For example, the Python protocol for a transformation experiment would be named `protocol-1.py`.
- **Protocol simulation**: `protocol-*.txt`. For the transformation example, this would be `protocol-1.txt`.
- **PDF instructions**: `protocol-*.pdf`. For the transformation example, this would be `protocol-1.pdf`.

Protocol 1: Heat Shock Transformations

Design

Use the JSON (protocol-1-config.json) and CSV (protocol-1-data.csv) files to create a protocol tailored for heat shock transformations. Define parameters as described in Table 1 and Table 2 below.

Parameter	Description	Example
`right_pipette_name`	Model of the pipette on the right mount (specify "NA" if not used)	`"p20_multi_gen2"`
`right_pipette_tiprack_name`	Tip rack type compatible with right pipette	`"opentrons_96_tiprack_20ul"`
`right_pipette_tiprack_slot`	Deck slot(s) where right pipette tip racks are placed	`[4, 5]`
`left_pipette_name`	Model of the pipette on the left mount (specify "NA" if not used)	`"p300_multi_gen2"`
`left_pipette_tiprack_name`	Tip rack type compatible with left pipette	`"opentrons_96_tiprack_300ul"`
`left_pipette_tiprack_slot`	Deck slot(s) where left pipette tip racks are placed	`[6]`
`dna_plate_name`	Labware containing DNA	`"opentrons_96_wellplate_200ul_pcr_full_skirt"`
`dna_plate_slot`	Deck slot of the labware containing DNA	`1`
`cells_plate_name`	Labware containing competent cells	`"opentrons_96_wellplate_200ul_pcr_full_skirt"`
`cells_plate_slot`	Deck slot of the labware containing competent cells	`2`
`media_plate_name`	Labware containing recovery media	`"usascientific_12_reservoir_22ml"`
`media_plate_slot`	Deck slot of the labware containing recovery media	`3`
`transformation_plate_name`	Labware where transformations occur (located in the thermocycler)	`"opentrons_96_wellplate_200ul_pcr_full_skirt"`
`pre_shock_incubation_temp`	Pre heat shock cooling incubation temperature ($^{\circ}C$)	`4`
`pre_shock_incubation_time_min`	Pre heat shock cooling incubation duration (minutes)	`30`
`heat_shock_temp`	Heat shock temperature ($^{\circ}C$)	`42`
`heat_shock_time_sec`	Heat shock duration (seconds)	`30`
`post_shock_incubation_temp`	Post heat shock incubation temperature ($^{\circ}C$)	`4`
`post_shock_incubation_time_min`	Post heat shock cooling incubation duration (minutes)	`2`
`recovery_temp`	Recovery incubation temperature ($^{\circ}C$)	`37`
`recovery_time_min`	Recovery incubation duration (minutes)	`60`

Table 1: JSON file configuration parameters for Protocol 1: Heat Shock Transformations (protocol-1-config.json).

Parameter	Description	Example
dna_id	Identifier for the DNA sample	pUC19
dna_source_well	Source well location on the DNA plate containing DNA sample (specify "NA" when adding no DNA)	A1
dna_volume	Volume of DNA to be transferred for transformation (specify 0 when adding no DNA) (μL)	1
cells_id	Identifier for the competent cells	DH5a
cells_source_well	Source well location on the cells plate containing the cells	A1
cells_volume	Volume of competent cells to be transformed (μL)	10
media_id	Identifier for the recovery media	SOC
media_source_well	Source well location on the media plate containing recovery media	A1
media_volume	Volume of media used for recovery (μL)	50
transformation_well	Destination well location on the transformation plate where the transformation mixture (DNA + cells + media) is assembled	A1

Table 2: CSV file configuration data for Protocol 1: Heat Shock Transformations (protocol-1-data.csv).

Example Preparation

Before running the protocol: **1.** Load the P20 8-Channel Pipette (right_pipette_name) on the right mount with a compatible Opentrons 96 Tiprack $20\,\mu L$ (right_pipette_tiprack_name) in slot 4 and 5 (right_pipette_tiprack_slot). Load the P300 8-Channel Pipette (left_pipette_name) on the left mount with a compatible tip rack (left_pipette_tiprack_name) in slot 6 (left_pipette_tiprack_slot) **2.** Prepare *Opentrons 96 Well* $200\,\mu L$ plate (dna_plate_name) with the plasmid DNA samples and place it on the deck slot 1 (dna_plate_slot). **3.** Prepare *Opentrons 96 Well* $200\,\mu L$ plate (cells_plate_name) with the chemically competent *E. coli* cells and keep the cells chilled on ice until needed and then place the plate on the deck slot 2 (cells_plate_slot). **4.** Prepare *USA Scientific 12 Well 22 mL* reservoir (media_plate_name) with the SOC recovery media and place it on the deck slot 3 (media_plate_slot). **5.** Prepare *Opentrons 96 Well* $200\,\mu L$ plate (transformation_plate_name) and place it into the Opentrons thermocycler. **Note:** If using the same labware for multiple reagents, simply repeat the labware name and deck slot in the configuration JSON file.

Example Execution

Step 1: APEX automatically selects which pipettes to use for each transfer, based on available pipettes and required volumes. If there is no pipette in one of the mounts or you don't want APEX to use an installed pipette, specify "NA" for that mount's pipette name parameter.

Step 2: APEX pre-cools the thermocycler to $4\,^{\circ}C$ (pre_shock_incubation_temp). Once the temperature is reached, the thermocycler opens.

Step 3: APEX gently resuspends the cells (cells_id) once per source well and distributes the cells from the cells plate wells A1-H1 (cells_source_well) across wells A1-H12 (transformation_well) on the transformation plate, with a volume of $10\,\mu L$/well (cells_volume). APEX will optimise by aspirating sufficient volume for multiple wells when possible and an additional disposal volume equal to the minimum pipette volume is aspirated to ensure accuracy. A single tip is used for all cells transfers, with a blow-out at the source well once finished.

Step 4: APEX transfers $1\,\mu L$/well (dna_volume) of DNA (dna_id) from wells A1-H12 (dna_source_well) on the DNA plate into A1-H12 (transformation_well) on the transformation plate, mixing gently once. A fresh tip is used for all DNA transfers, and after a blowout, the pipette descends again to add any droplet left at the tip to the well.

Step 5: The thermocycler closes and incubates the cells at $4\,°C$ (pre_shock_incubation_temp) for 30 minutes (pre_shock_incubation_time_min), followed by a heat shock at $42\,°C$ (heat_shock_temp) for 30 seconds (heat_shock_time_sec), and an incubation at $4\,°C$ (post_shock_incubation_temp) for 2 minutes (post_shock_incubation_time_min). APEX automatically calculates the maximum reaction volume based on the content of the well, and configures the thermocycler block accordingly.

Step 6: Thermocycler opens again, and APEX transfers $50\,\mu L$/well (media_volume) of SOC media (media_id) from wells A1-H1 (media_source_well) on the media plate into the wells A1-H12 (destination_well) on the transformation plate, resuspending the cells 3 times with the mixing volume calculated by APEX as half the total volume of all the contents per well (capped at the pipette's maximum volume).

Step 7: Finally, the thermocycler closes and heats up to $37\,°C$ (recovery_temp) and incubates the cells for 60 minutes (recovery_time_min). Once finished, the cells are ready for selection via spotting on the selective media plate (APEX Protocol 2)

Protocol 2: Colony Selection

Design

Use the JSON (protocol-2-config.json) and CSV (protocol-2-data.csv) files to create a protocol tailored for colony selection via selective agar plate spotting. Define parameters as described in Table 3 and Table 4 below.

Parameter	Description	Example
`pipette_name`	Model of the pipette	`"p20_multi_gen2"`
`pipette_mount`	Side of the robot the pipette is mounted on	`"left" or "right"`
`tiprack_name`	Tip rack type compatible with pipette	`"opentrons_96_tiprack_20ul"`
`tiprack_slot`	Deck slot(s) where pipette tip racks are placed	`[4]`
`transformation_plate_name`	Labware containing the transformations	`"opentrons_96_wellplate_200ul_pcr_full_skirt"`
`agar_plate_name`	Labware of the agar plate	`"nunc96grid_96_wellplate_10ul"`
`agar_plate_area`	The base area of the agar plate (cm^2)	`9.4692`
`empty_plate_weight`	The weight of the plate without agar (g)	`38.92`
`agar_density`	Agar density (g cm^{-3})	`0.911`
`mixing_volume`	Volume of resuspension mixing of the transformed cells before aspirating (µL)	`1`
`additional_volume`	Additional aspiration volume ensures accurate dispensing (that volume will not be dispensed onto the plate) (µL)	`1`
`spotting_height`	Z-axis offset from agar surface for dispensing (positive values increase height; negative values decrease height; at value 0 the pipette tip will be right at the height of the agar) (mm)	`0.5`

Table 3: JSON file configuration parameters for Protocol 2: Colony Selection (protocol-2-config.json).

Parameter	Description	Example
`sample_id`	Identifier for the transformation sample	`pUC19-DH5a`
`agar_plate_slot`	Deck slot(s) of the agar plate(s)	`1`
`agar_plate_weight`	The weight of the plate(s) with agar (g)	`68.92`
`transformation_source_well`	Source well location of the transformed cells	`A1`
`spotting_volume`	Volume of cells to spot onto agar (µL)	`5`
`spotting_well`	Destination well location on the agar plate where the transformed cells are spotted	`A1`

Table 4: CSV file configuration data for Protocol 2: Colony Selection (protocol-2-data.csv).

Example Preparation

Before running the protocol: **1.** Load the P20 8-Channel Pipette (pipette_name) on the right mount (pipette_mount) with a compatible Opentrons 96 Tiprack $20\,\mu L$ (tiprack_name) in slot 4 (tiprack_slot). **2.** Prepare *Opentrons 96 Well* $200\,\mu L$ plate (transformation_plate_name) with the transformed cells and place it the Opentrons thermocycler. **3.** Prepare *Nunc OmniTray Single-Well* plate(s) (agar_plate_name) with LB agar containing the appropriate selection markers and wait for the agar to solidify and dry. Place it on the deck slot 1 (agar_plate_slot). **4.** Record the weights of the plate(s) before (empty_agar_plate_weight) and after (agar_plate_weight) pouring molten LB agar. **Note:** Proper drying of plates is essential to prevent excessive diffusion of the spotted cells.

Example Execution

Step 1: APEX calculates the agar height based on the plate dimensions (agar_plate_area), plate weights (empty_agar_plate_weight, agar_plate_weight), and agar density (agar_density). The user can adjust the spotting height parameter (spotting_height) to modify the height of spotting relative to the agar surface. It is recommended to verify the precise plate well depth and agar height using the provided Jupyter notebook.

Step 2: APEX picks up a new tip for each transfer and resuspends the cells 3 times before aspiration with the mixing volume of $20\,\mu L$ (mixing_volume). APEX aspirates $5\,\mu L$ (spotting_volume) of the transformed cells (sample_id) from wells A1-H12 (transformation_source_well) on the transformation plate and transfers them to the A1-H12 wells (spotting_well) on the agar plates. For each transfer an additional volume of $1\,\mu L$ (additional_volume) is aspirated that will not be dispensed. This approach is preferred over using a blowout to prevent sample dispersion and to achieve more precise volume dispensation. After dispensing, the protocol includes a 5-second delay to ensure smooth transfer of the sample.

Step 3: To spot the same sample multiple times without changing tips (technical replicates), multiple destination wells can be listed in the destination_well field using the | symbol. For instance, `A1|A2` directs APEX to dispense aliquots from the source well into both wells A1 and A2.

Step 4: Allow the spotted plates to dry before removing them from the OT-2 deck. Transfer the plates for incubation at $37\,°C$ for 12-18 hours. After incubation, when colonies have formed, the plates are ready for colony sampling using APEX Protocol 3 (Colony Sampling).

Protocol 3: Colony Sampling

Design

Use the the JSON (protocol-3-config.json) and CSV (protocol-3-data.csv) files to create a protocol tailored for colony sampling. Define parameters as described in Table 5 and Table 6 below.

Parameter	Description	Example
`right_pipette_name`	Model of the pipette on the right mount (specify "NA" if not used)	`"p20_multi_gen2"`
`right_pipette_tiprack_name`	Tip rack type compatible with right pipette	`"opentrons_96_tiprack_20ul"`
`right_pipette_tiprack_slot`	Deck slot(s) where right pipette tip racks are placed	`[4]`
`left_pipette_name`	Model of the pipette on the left mount (specify "NA" if not used)	`"p300_multi_gen2"`
`left_pipette_tiprack_name`	Tip rack type compatible with left pipette	`"opentrons_96_tiprack_300ul"`
`left_pipette_tiprack_slot`	Deck slot(s) where left pipette tip racks are placed	`[5]`
`media_plate_name`	Labware containing media	`"usascientific_12_reservoir_22ml"`
`media_plate_slot`	Deck slot of the labware containing media	`1`
`culture_plate_name`	Labware where cells will be propagated	`"usascientific_96_wellplate_2.4ml_deep"`
`culture_plate_slot`	Deck slot of the culture plate	`2`
`agar_plate_name`	Labware of the agar plate with devleoped colonies	`"nunc96grid_96_wellplate_10ul"`
`agar_plate_area`	The base area of the agar plate (cm^2)	`9.4692`
`empty_plate_weight`	The weight of the empty plate without agar (g)	`38.92`
`agar_density`	Agar density (g cm^{-3})	`0.911`
`sampling_method`	Method used for sampling colonies from the agar plate	`"spiral"`
`sampling_height`	Z-axis offset from agar surface for dispensing (positive values increase height; negative values decrease height; at value 0 the pipette tip will be right at the height of the agar) (mm)	`-0.5`
`spiral_radius`	Maximum radius of the spiral sampling pattern (mm)	`2.5`
`spiral_points`	Number of discrete sampling points in the spiral path	`25`
`spiral_rotations`	Number of complete turns in the spiral sampling pattern	`3`

Table 5: JSON configuration data details for Protocol 3: Colony Sampling (protocol-3-config.json.)

Example Preparation

Before running the protocol: **1.** Load the P20 8-Channel Pipette (right_pipette_name) on the right mount with a compatible Opentrons 96 Tiprack $20\,\mu L$ (right_pipette_tiprack_name) in slot 4 (right_pipette_tiprack_slot). Load the P300 8-Channel Pipette (left_pipette_name) on the left mount with a compatible tip rack (left_pipette_tiprack_name) in slot 5 (left_pipette_tiprack_slot) **2.** Prepare

Parameter	Description	Example
`colony_id`	Identifier for the bacterial colonies on the agar plate	`pUC19-BL21(DE3)`
`agar_plate_slot`	Deck slot where the agar plate with colonies is placed	`3`
`colony_source_well`	Source well location on the agar plate from which to sample colonies	`A1`
`media_id`	Identifier of the media	`LB-kan`
`media_source_well`	Source well location of the recovery media on the media plate	`A3`
`media_volume`	Volume of media for culturing (µL)	`500`
`culture_well`	Destination well location on the culture plate where sampled colonies will be inoculated into fresh media	`A1`

Table 6: CSV file configuration data for Protocol 3: Colony Sampling (protocol-3-data.csv).

USA Scientific 12 Well 22 mL reservoir (media_plate_name) with the LB media and place it on the deck slot 1 (media_plate_slot). **3.** Prepare *USA Scientific 96 Deep Well 2.4 mL* plate (culture_plate_name) and place it on the deck slot 2 (culture_plate_slot). **4.** Prepare *Nunc OmniTray Single-Well* plate(s) (agar_plate_name) with LB agar with developed colonies and place it on the deck slot 3 (agar_plate_slot). Record the weights of the plates(s) with and without LB agar. **5.** Decide on a colony sampling strategy (”spiral” or ”pierce”).

Example Execution

Step 1: APEX automatically selects which pipettes to use for each transfer, based on available pipettes and required volumes. If there is no pipette in one of the mounts or you don't want APEX to use an installed pipette, specify ”NA” for that mount's pipette name parameter.

Step 2: APEX distributes $500\,\mu\mathrm{L}$/well (media_volume) of growth media supplemented with appropriate antibiotics (media_id) from wells A1-H3 (media_source_well) from the media plate into wells A1-H12 (culture_well) on the culture plate. APEX will optimise by aspirating sufficient volume for multiple wells when possible and an additional disposal volume equal to the minimum pipette volume is aspirated to ensure accuracy. A single tip is used for all media transfers, with a blow-out at the source well once finished.

Step 3: Based on the selected sampling_method, APEX performs colony sampling:

- For ”spiral” method: APEX moves the pipette to the center of the specified well containing colonies (colony_id in colony_source_well) and follows a spiral path (defined by spiral_radius, spiral_points, and spiral_rotations) to sample the colonies at sampling_height.
- For ”single pierce” method: APEX moves to the center of the specified well and pierces the agar once at sampling_height.

Step 4: APEX transfers the sampled cells into the media in the culture plate (culture_well) and mixes the media twice with the mixing volume calculated by APEX as half the total volume of the media volume per well (capped at the pipette's maximum volume).

Step 5: After sampling all selected wells with colonies, the culture plate is ready to be moved for shaking incubation at $37\,^{\circ}\mathrm{C}$. After incubation, the plates are ready for protein expression induction using APEX Protocol 4 (Protein Expression Induction).

Protocol 4: Protein Expression Induction

Design

Use the JSON (protocol-4-config.json) and CSV (protocol-4-data.csv) files to create a protocol tailored for protein expression induction. Define parameters as described in Table 7 and Table 8 below.

Parameter	**Description**	**Example**
`right_pipette_name`	Model of the pipette on the right mount (specify "NA" if not used)	`"p20_multi_gen2"`
`right_pipette_tiprack_name`	Tip rack type compatible with right pipette	`"opentrons_96_tiprack_20ul"`
`right_pipette_tiprack_slot`	Deck slot(s) where right pipette tip racks are placed	`[5,6]`
`left_pipette_name`	Model of the pipette on the left mount (specify "NA" if not used)	`"p300_multi_gen2"`
`left_pipette_tiprack_name`	Tip rack type compatible with left pipette	`"opentrons_96_tiprack_300ul"`
`left_pipette_tiprack_slot`	Deck slot(s) where left pipette tip racks are placed	`[7]`
`media_plate_name`	Labware containing media	`"usascientific_12_reservoir_22ml"`
`media_plate_slot`	Deck slot of the labware containing media	`1`
`culture_plate_name`	Labware where cells will be propagated	`"usascientific_96_wellplate_2.4ml_deep"`
`culture_plate_slot`	Deck slot of the culture plate	`2`
`inducer_plate_name`	Labware containing the chemical inducer	`"opentrons_96_wellplate_200ul_pcr_full_skirt"`
`inducer_plate_slot`	Deck slot of the inducer plate	`4`
`expression_plate_name`	Labware were the fresh culture will be propagated and the protein expression will be induced	`"usascientific_96_wellplate_2.4ml_deep"`
`expression_plate_slot`	Deck slot of the expression plate	`3`

Table 7: JSON configuration data details for Protocol 4: Protein Expression Induction (protocol-4-config.json).

Example Preparation

Before running the protocol: **1.** Load the P20 8-Channel Pipette (right_pipette_name) on the right mount with a compatible Opentrons 96 Tiprack $20\,\mu\mathrm{L}$ (right_pipette_tiprack_name) in slot 5 and 6 (right_pipette_tiprack_slot). Load the P300 8-Channel Pipette (left_pipette_name) on the left mount with a compatible tip rack (left_pipette_tiprack_name) in slot 7 (left_pipette_tiprack_slot) **2.** Prepare *USA Scientific 12 Well 22 mL* Reservoir (media_plate_name) with the media supplemented with antibiotics and place it on the deck slot 1(media_plate_slot). **3.** Prepare *USA Scientific 96 Deep Well 2.4 mL* plate (culture_plate_name) with the cultures and place it on the deck slot 2 (culture_plate_slot). **4.** Prepare *Opentrons 96 Well* $200\,\mu\mathrm{L}$ plate (inducer_plate_name) with the chemical inducer (e.g. IPTG)

Parameter	Description	Example
`culture_id`	Identifier for the culture sample	`pUC19-BL21(DE3)`
`culture_source_well`	Source well location on the culture plate containing overnight cultures	`A1`
`culture_volume`	Volume of culture to be propagated in fresh media (μL)	`5`
`media_id`	Identifier of media	`fresh-LB`
`media_source_well`	Source well location on the media plate containing media	`A1`
`media_volume`	Volume of media to be distributed per well into the expression plate (μL)	`500`
`inducer_id`	Identifier of the chemical inducer	`0.1mM-IPTG`
`inducer_source_well`	Source well location on the inducer plate (specify "NA" when adding no inducer)	`A1`
`inducer_volume`	Volume of the chemical inducer to be transferred per well into the expression plate (μL) (specify 0 when adding no inducer)	`5`
`expression_well`	Destination well location on expression plate where cells will be propagated and protein expression will be induced	`A1`

Table 8: CSV file configuration data for Protocol 4: Protein Expression Induction (protocol-4-data.csv).

and place it on the deck slot 4 (inducer_plate_slot). **5.** Prepare *USA Scientific 96 Deep Well 2.4 mL* plate (expression_plate_name) as an expression plate where fresh culture will be propagated and protein expression will be induced and place it on the deck slot 3 (expression_plate_slot). **Note:** If using the same labware for multiple reagents, simply repeat the labware name and deck slot in the configuration JSON file.

Example Execution

Step 1: APEX automatically selects which pipettes to use for each transfer, based on available pipettes and required volumes. If there is no pipette in one of the mounts or you don't want APEX to use an installed pipette, specify "NA" for that mount's pipette name parameter.

Step 2: APEX distributes $500\,\mu L$/well (media_volume) of LB media supplemented with appropriate antibiotics (media_id) from wells A1-H3 (media_source_well) from the media plate into the wells A1-H12 (expression_well) on the expression plate.

Step 3: APEX aliquots $5\,\mu L$ (culture_volume) of overnight cultures from wells A1-H12 (culture_source_well) on the culture plate using a new tip each time and inoculates the fresh media in wells A1-H12 (expression_well) on the expression plate by mixing it twice with the mixing volume calculated by APEX as half the total volume of media per well (capped at the pipette's maximum volume).

Step 4: The protocol pauses to allow for incubation. Incubate the plate with shaking at $37\,°C$ and grow culture to desired OD_{600}.

Step 5: Upon reaching the desired OD_{600}, resume the protocol. APEX adds $5\,\mu L$/well (inducer_volume) of IPTG (inducer_id) from wells A1-H1 (inducer_well) on the inducer plate to each culture in wells A1-H12 (expression_well) on the expression plate to induce the protein expression.

Step 6: Once the protocol is finished, continue the incubation of the cultures with shaking at $20\,°C$ to facilitate protein production.

Protocol 5: Colony PCR Sample Preparation

Design

Use the JSON (protocol-5-config.json) and CSV (protocol-5-data.csv) files to create a protocol tailored for colony PCR sample preparation. Define parameters as described in Table 9 and Table 10 below.

Parameter	Description	Example
`right_pipette_name`	Model of the pipette on the right mount (specify ”NA” if not used)	`"p20_multi_gen2"`
`right_pipette_tiprack_name`	Tip rack type compatible with right pipette	`"opentrons_96_tiprack_20ul"`
`right_pipette_tiprack_slot`	Deck slot(s) where right pipette tip racks are placed	`[5]`
`left_pipette_name`	Model of the pipette on the left mount (specify ”NA” if not used)	`"p300_multi_gen2"`
`left_pipette_tiprack_name`	Tip rack type compatible with left pipette	`"opentrons_96_tiprack_300ul"`
`left_pipette_tiprack_slot`	Deck slot(s) where left pipette tip racks are placed	`[6]`
`reagent_plate_name`	Labware containing PCR reagents (master mix, primers, water)	`"opentrons_96_wellplate_200ul_pcr_full_skirt"`
`reagent_plate_slot`	Deck slot where reagent plate is placed	1
`pcr_plate_name`	Labware where PCR reactions will be assembled (placed in the thermocycler)	`"opentrons_96_wellplate_200ul_pcr_full_skirt"`
`mmastermix_mix_number`	Number of mixing of mastermix with the colony solution	4
`amplification_cycles`	Number of thermal cycling repetitions	30
`lid_temp`	Temperature of thermocycler lid (°C)	105
`initial_denaturation_temp`	Temperature for initial denaturation step (°C)	94
`initial_denaturation_time_min`	Duration of initial denaturation (minutes)	3
`denaturation_temp`	Temperature for cycle denaturation step (°C)	94
`denaturation_time_sec`	Duration of cycle denaturation (seconds)	30
`annealing_temp`	Temperature for primer annealing step (°C)	55
`annealing_time_sec`	Duration of primer annealing (seconds)	30
`extension_temp`	Temperature for DNA extension step (°C)	68
`extension_time_sec`	Duration of DNA extension (seconds)	60
`final_extension_temp`	Temperature for final extension step (°C)	68
`final_extension_time_min`	Duration of final extension (minutes)	5
`hold_temp`	Final holding temperature (°C)	4

Table 9: JSON file configuration parameters for Protocol 5: Colony PCR Sample Preparation Protocol (protocol-5-config.json).

Example Preparation

Before running the protocol: **1.** Load the P20 8-Channel Pipette (right_pipette_name) on the right mount with a compatible Opentrons 96 Tiprack $20\,\mu L$ (right_pipette_tiprack_name)

Parameter	Description	Example
colony_id	Identifier for the colony for PCR analysis	sample_1
water_source_well	Source well location in the reagent plate containing nuclease-free water for colony resuspension	A1
water_volume	Volume of nuclease-free water to be dispensed for colony resuspension (µL)	11.5
master_mix_source_well	Source well location in the reagent plate containing the PCR master mix	A2
master_mix_volume	Volume of PCR master mix (µL)	13.5
pcr_well	Destination well location in the PCR plate where the PCR reaction will occur	A1

Table 10: CSV file configuration data for Protocol 5: Colony PCR Sample Preparation Protocol (protocol-5-data.csv).

in slot 5 (right_pipette_tiprack_slot). Load the P300 8-Channel Pipette (left_pipette_name) on the left mount with a compatible tip rack (left_pipette_tiprack_name) in slot 6 (left_pipette_tiprack_slot) **2.** Prepare *Opentrons 96 Well* $200\,\mu\mathrm{L}$ plate (reagent_plate_name) with the PCR master mix and nuclease-free water and place it on the deck slot 2 (reagent_plate_slot). **3.** Prepare *Opentrons 96 Well* $200\,\mu\mathrm{L}$ plate (pcr_plate_name) with where the PCR raction will occur and place it in the Opentrons thermocycler. Ensure the thermocycler is turned on and available. **4.** Prepare plates with bacterial colonies ready for processing. **5.** Prepare toothpicks or tips for manual colony picking. **Note:** If using the same labware for multiple reagents, simply repeat the labware name and deck slot in the configuration JSON file.

Example Execution

Step 1: APEX automatically selects which pipettes to use for each transfer, based on available pipettes and required volumes. If there is no pipette in one of the mounts or you don't want APEX to use an installed pipette, specify "NA" for that mount's pipette name parameter.

Step 2: APEX distributes $11.5\,\mu\mathrm{L}$/well (water_volume) of nuclease-free water from wells A1-H1 (water_source_well) on the reagent plate across the wells A1-H12 (pcr_well) on the PCR plate. APEX will optimise by aspirating sufficient volume for multiple wells when possible and an additional disposal volume equal to the minimum pipette volume is aspirated to ensure accuracy. A single tip is used for all cells transfers, with a blow-out at the source well once finished.

Step 3: APEX pauses for manual intervention. The user removes the PCR plate to pick individual bacterial colonies and resuspend them in the distributed water. The user places the plate into the thermocycler module and resumes the protocol.

Step 4: APEX transfers $13.5\,\mu L$ (mastermix_volume) of the PCR master mix from the well A2-H2 (mastermix_source_well) on the reagent plate into each well containing resuspended colonies (pcr_well) on the PCR plate. The contents of each well are mixed 4 times (mastermix_mix_number) with the mixing volume calculated by APEX as half the total volume of master mix and water per well (capped at the pipette's maximum volume). A new tip is used for each transfer with blow-out at the destination well.

Step 5: The thermocycler closes its lid and sets lid temperature to $105\,^{\circ}C$ (lid_temp). Initial denaturation is performed at $94\,^{\circ}C$ (initial_denaturation_temp) for 3 minutes (initial_denaturation_time_min). The PCR thermal cycling then begins, with 30 repetitions (amplification_cycles) of: denaturation at $94\,^{\circ}C$ (denaturation_temp) for 30 seconds (denaturation_time_sec), primer annealing at $55\,^{\circ}C$ (annealing_temp) for 30 seconds (annealing_time_sec), and DNA extension at $68\,^{\circ}C$ (extension_temp) for 60 seconds (extension_time_sec). APEX automatically calculates the maximum reaction volume based on the combined PCR master mix, and colony suspension in water volumes, and configures the thermocycler block accordingly. After cycling completes, a final extension at $68\,^{\circ}C$ (final_extension_temp) for 5 minutes (final_extension_time_min). Samples are then held at $4\,^{\circ}C$ (hold_temp) until retrieval and are ready for subsequent analysis via gel electrophoresis.

Protocol 6: SDS PAGE Sample Preparation

Design

Use the JSON (protocol-6-config.json) and CSV (protocol-6-config.csv) files to create a protocol tailored for SDS PAGE sample preparation. Define parameters as described in Table 11 and Table 12 below.

Parameter	Description	Example
right_pipette_name	Model of the pipette on the right mount (specify "NA" if not used)	"p20_multi_gen2"
right_pipette_tiprack_name	Tip rack type compatible with right pipette	"opentrons_96_tiprack_20ul"
right_pipette_tiprack_slot	Deck slot(s) where right pipette tip racks are placed	[5]
left_pipette_name	Model of the pipette on the left mount (specify "NA" if not used)	"p300_multi_gen2"
left_pipette_tiprack_name	Tip rack type compatible with left pipette	"opentrons_96_tiprack_300ul"
left_pipette_tiprack_slot	Deck slot(s) where left pipette tip racks are placed	[6]
culture_plate_name	Labware containing culture	"usascientific_96_wellplate_2.4ml_deep"
culture_plate_slot	Deck slot where culture plate is placed	1
reagent_plate_name	Labware containing reagents (lysis buffer, SDS buffer)	"usascientific_12_reservoir_22ml"
reagent_plate_slot	Deck slot where reagent plate is placed	2
lysate_plate_name	Labware to which lysate will be transfered	"usascientific_12_reservoir_22ml"
lysate_plate_slot	Deck slot where lysate plate is placed	3
destination_plate_name	Labware used for SDS sample prep (located in the thermocycler)	"opentrons_96_wellplate_200ul_pcr_full_skirt"
culture_mix_number	Number of mixing repetitions before culture transfer	3
culture_mix_volume	Volume for mixing culture (µL)	20

Table 11: JSON file configuration parameters for Protocol 6: SDS PAGE Sample Preparation (protocol-6-config.json).

Parameter	Description	Example
sample_id	Identifier for the sample	sample_1
culture_source_well	Source well location in the culture plate containing bacterial culture	A1
culture_volume	Volume of bacterial culture to be analysed for expression (µL)	150
lysis_buffer_source_well	Source well location in the reagent plate containing lysis buffer	A1
lysis_buffer_volume	Volume of lysis buffer (µL)	40
sds_buffer_source_well	Source well location in the reagent plate containing SDS buffer	A1
sds_buffer_volume	Volume of SDS buffer (µL)	10
destination_well	Destination well location in the destination plate where the SDS sample prep will occur	A1

Table 12: CSV file configuration data for Protocol 6: SDS PAGE Sample Preparation (protocol-6-data.csv).

Example Preparation

Before running the protocol: **1.** Load the P20 8-Channel Pipette (right_pipette_name) on the right mount with a compatible Opentrons 96 Tiprack $20\,\mu\mathrm{L}$ (right_pipette_tiprack_name) in slot 5 (right_pipette_tiprack_slot). Load the P300 8-Channel Pipette (left_pipette_name) on the left mount with a compatible tip rack (left_pipette_tiprack_name) in slot 6 (left_pipette_tiprack_slot) **2.** Prepare *USA Scientific 96 Deep Well 2.4 mL* plate (culture_plate_name) with the bacterial cultures and place it on the deck slot 1 (culture_plate_slot). **3.** Prepare *USA Scientific 12 Well 22 mL* reservoir (reagent_plate_name) with the reagents (lysis buffer and SDS buffer) and place it on the deck slot 2 (reagent_plate_slot). **4.** Prepare *USA Scientific 96 Deep Well 2.4 mL* plate (lysate_plate_name) where lysate will be transferred and place it on the deck slot 3(culture_plate_slot). **5.** Prepare *Opentrons 96 Well* $200\,\mu\mathrm{L}$ plate (destination_plate_name) where the SDS sample prep will occur and place it on the int he Opentrons thermocycler. Ensure the thermocycler is turned on and available. **6.** Set up the external centrifuge with plate rotors. **Note:** If using the same labware for multiple reagents, simply repeat the labware name and deck slot in the configuration JSON file.

Example Execution

Step 1: APEX automatically selects which pipettes to use for each transfer, based on available pipettes and required volumes. If there is no pipette in one of the mounts or you don't want APEX to use an installed pipette, specify "NA" for that mount's pipette name parameter.

Step 2: APEX transfers $150\,\mu\mathrm{L}$ (culture_volume) of bacterial cultures from the wells A1-H12 (culture_source_well) on the culture culture plate to the wells A1-H12 (destination_well) on the lysis plate. Before aspiration, APEX mixes the cultures 3 times (culture_mix_number) with a volume of $100\,\mu\mathrm{L}$ (culture_mix_volume) to resuspend the cells. A new tip is used for each transfer with blow-out at the destination well.

Step 3: The protocol pauses with a message prompting the user for manual centrifugation of the lysis plate. After centrifugation and returning the plate to the deck, protocol is resumed and pipetting parameters are automatically adjusted to avoid disturbing the cell pellet. The pipette's well bottom clearance is set to $0.5\,\mathrm{mm}$ (pellet_well_bottom_clearance) which can be adjusted based on the pellet size. Aspiration flow rate is reduced to $25\,\%$ (supernatant_flow_rate_aspirate) for gentle supernatant removal. The supernatant is discarded into the waste container, and pipette settings return to default values.

Step 4: APEX transfers $40\,\mu L$/well (lysis_buffer_volume) of the lysis buffer from wells A1-H1 (lysis_buffer_source_well) on the reagent plate to each well containing the cell pellets (destination_well). APEX performs 5 mixes (lysis_mix_number) to thoroughly resuspend the pellet. Mixing volume is calculated by APEX as half the lysis buffer volume per well (capped at the maximum volume of the pipette). A new tip is used for each transfer with blow-out at the destination well.

Step 5: Samples are incubated for $30\,\mathrm{min}$ (lysis_incubation_time_min) at room temperature. The protocol then pauses for manual centrifugation of the lysis plate to pellet cell debris.

Step 6: APEX transfers $40\,\mu L$ (lysis_buffer_volume) of the lysate from wells A1-H12 (destination_well) on the lysate plate into the wells A1-H12 (destination_well) on the destination plate located in the thermocycler. A new tip is used for each transfer with blow-out at the destination well.

Step 7: APEX transfers $10\,\mu L$/well (sds_buffer_volume) of the 5x SDS buffer from wells A2-H12 (sds_buffer_source_well) on the reagent plate to each well A1-H12 (destination_well) containing the lysate. APEX performs 4 mixes (sds_mix_number). Mixing volume is calculated as half the total volume of lysate and SDS buffer in the well (capped at the maximum volume of the pipette). A new tip is used for each transfer with a blow-out at the destination well.

Step 8: The thermocycler closes its lid and sets the lid temperature to $105\,^{\circ}\mathrm{C}$ (lid_temp). The maximum reaction volume is calculated by APEX and thermocycler block max volume is set accordingly. Samples are heated to $95\,^{\circ}\mathrm{C}$ (denaturation_temp) for $5\,\mathrm{min}$ (denaturation_time_min). The thermocycler deactivates both lid and block heater once finished. The denatured protein samples are now ready for gel loading.

Protocol 7: OD_{600} Reading Sample Preparation

Design

Use the JSON (protocol-6-config.json) and CSV (protocol-7-config.csv) files to create a protocol tailored for OD_{600} reading sample preparation. Define parameters as described in 13 and 14 below.

Parameter	Description	Example
`right_pipette_name`	Model of the pipette on the right mount (specify "NA" if not used)	`"p20_multi_gen2"`
`right_pipette_tiprack_name`	Tip rack type compatible with right pipette	`"opentrons_96_tiprack_20ul"`
`right_pipette_tiprack_slot`	Deck slot(s) where right pipette tip racks are placed	`[5]`
`left_pipette_name`	Model of the pipette on the left mount (specify "NA" if not used)	`"p300_multi_gen2"`
`left_pipette_tiprack_name`	Tip rack type compatible with left pipette	`"opentrons_96_tiprack_300ul"`
`left_pipette_tiprack_slot`	Deck slot(s) where left pipette tip racks are placed	`[6]`
`diluent_plate_name`	Labware containing diluent (e.g. water, media)	`"corning_96_wellplate_360ul_flat"`
`diluent_plate_slot`	Deck slot where diluent plate is placed	`1`
`culture_plate_name`	Labware containing culture	`"usascientific_96_wellplate_2.4ml_deep"`
`culture_plate_slot`	Deck slot where culture plate is placed	`2`
`reading_plate_name`	Labware for reading OD_{600} of the cultures	`"opentrons_96_wellplate_200ul_pcr_full_skirt"`
`reading_plate_slot`	Deck slot where reading plate is placed	`3`
`culture_mix_number`	Number of mixing of bacterial culture before transfer	`3`
`culture_mix_volume`	Volume for mixing bacterial culture (µL)	`20`

Table 13: JSON file configuration parameters for Protocol 7: OD_{600} Reading Sample Preparation (protocol-7-config.json).

Parameter	Description	Example
`culture_id`	Identifier for the sample	`sample_1`
`diluent_source_well`	Source well location in the diluent plate containing diluent	`A1`
`diluent_volume`	Volume of diluent to be dispensed for diluting cultures (µL)	`90`
`culture_source_well`	Source well location in the culture plate containing bacterial culture (specify "NA" when adding no culture for the blank))	`A1`
`culture_volume`	Volume of bacterial culture to be transferred (specify 0 when adding no culture for the blank) (µL)	`10`
`reading_well`	Destination well position in the reading plate where the bacterial culture will be diluted for OD_{600} measurement	`A1`

Table 14: CSV file configuration data for Protocol 7: OD_{600} Reading Sample Preparation (protocol-7-data.csv).

Example Preparation

Before running the protocol: **1.** Load the P20 8-Channel Pipette (right_pipette_name) on the right mount with a compatible Opentrons 96 Tiprack $20\,\mu\mathrm{L}$ (right_pipette_tiprack_name) in slot 5 (right_pipette_tiprack_slot). Load the P300 8-Channel Pipette (left_pipette_name) on the left mount with a compatible tip rack (left_pipette_tiprack_name) in slot 6 (left_pipette_tiprack_slot) **2.** Prepare *USA Scientific 12 Well 2.4 mL* plate (culture_plate_name) with the cultures and place it on the deck slot q (culture_plate_slot). **3.** Prepare *Corning 96 Well* $360\,\mu\mathrm{L}$ plate (reading_plate_name) and place it on the deck slot 3 (reading_plate_slot). **4.** Prepare USA Scientific 12 Well 22 mL Reservoir (diluent_plate_name) with the diluent (typically water or media) and place it on the deck slot 2 (diluent_plate_slot). **Note:** If using the same labware for multiple reagents, simply repeat the labware name and deck slot in the configuration JSON file.

Example Execution

Step 1: APEX automatically selects which pipettes to use for each transfer, based on available pipettes and required volumes. If there is no pipette in one of the mounts or you don't want APEX to use an installed pipette, specify "NA" for that mount's pipette name parameter.

Step 2: APEX distributes $90\,\mu\mathrm{L}$/well (diluent_volume) of diluent from wells A1-H1 (diluent_source_well) on the diluent plate across the wells A1-H11 (reading_well) and $100\,\mu\mathrm{L}$/well (diluent_volume + culture_volume) in wells A12-H12. APEX will optimise by aspirating sufficient volume for multiple wells when possible and an additional disposal volume equal to the minimum pipette volume is aspirated to ensure accuracy. A single tip is used for all diluent transfers, with a blow-out at the source well once finished.

Step 3: APEX transfers $10\,\mu\mathrm{L}$ (culture_volume) of bacterial cultures from wells A1-H12 (culture_source_well) on the culture plate to the wells A1-H11 (reading_well) on the reading plate. Before aspiration, APEX mixes the cultures 5 times (culture_mix_number) with a volume of $250\,\mu\mathrm{L}$ (culture_mix_volume) to resuspend the cells. A new tip is used for each transfer with a blow-out at the destination well.

Step 4: For calibration blanks, specify "NA" in the culture_source_well field and "0" in the culture_source_volume. For blank samples, set diluent_volume equal to the total desired final volume. Once the protocol is finished, samples can be analysed using a plate reader to determine bacterial cell density.